I'm Diya Shrivastava.
I build machine-learning systems people can rely on.

Models that know how confident they are, and AI assistants that show their sources — or say "I don't know." I care about work that's useful beyond the demo.

4× Microsoft Certified · Applied ML & GenAI Developer · Ahmedabad, India

- Open to AI/ML and Data Science internships — available now.

See my work Download résumé

Email GitHub LinkedIn

What I'm doing these days.

- Third year of B.Tech CSE at Lovely Professional University, currently studying Design & Analysis of Algorithms and DevOps.
- Practising data structures and algorithms in Java on Codeforces.

Updated November 2026

Latest problems I've worked on.

Legal & procurement · 2026

ClauseCheck

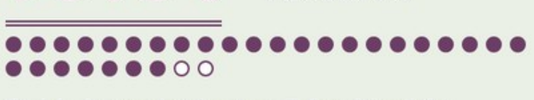
Procurement RAG Assistant

Procurement teams lose hours hunting through supplier contracts. ClauseCheck answers their questions with the exact clause cited — and refuses when the contracts don't say.

Python · ChromaDB · BM25 · FastAPI · Streamlit

Read case study →

93.5% recall@5



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Payments & fintech risk · 2026

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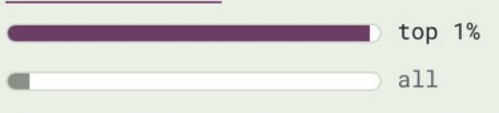
16.2× lift



Dispute rate in the riskiest 1% (66 of 452) against 0.90% across all transactions.

`₹11.99L net savings` — 39.1% of exposure

Product analytics · 2026 · Team project

StreakForge

Fitness Engagement & Retention Insight Platform

Fitness apps can see who stops showing up, but not why. StreakForge shows which engagement habits keep members coming back — and finds the biggest leak is members who never get started.

Python · Pandas · Plotly · Streamlit

Read case study →

81.2% drop-off

Only 18.8% of members activate in their first 7 days.

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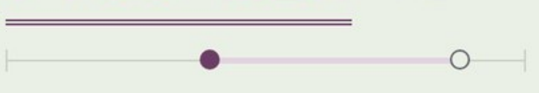
7.83 min MAE



Model (filled) against the 17.48-min baseline (open), on a 0–20 min scale.

`55% better` — than the baseline

All projects →

How I try to work.

> *Honest numbers.*
> Every metric shown with its baseline and its limits.

> *Decisions, not just predictions.*
> A model is only useful when its output is something someone can act on.

> *Work that lasts beyond the demo.*
> Tests, documentation, and saying what's unfinished.

How I approach a problem.

1
Understand the decision
Who will act on the output, and what a mistake costs them.

→

2
Respect the data
Clean it, question it, and split it the way time actually flows.

→

3
Evaluate honestly
Baselines, calibration, and the cases where it fails.

→

4
Ship something usable
Tested, documented, and clear about its limits.

Major certifications

Microsoft certifications earned alongside my degree in 2026. Each one links to Microsoft's own verification page.

`pl-200.pdf`
landscape certificate

Power Platform Functional Consultant Associate

Microsoft · PL-200 · Aug 2026

View certificate Verify →

`pl-900.pdf`
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Power Platform Fundamentals

Microsoft · PL-900 · Aug 2026

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`ai-900.pdf`
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Azure AI Fundamentals

Microsoft · AI-900 · Feb 2026

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GitHub Foundations

Microsoft · GH-900 · Feb 2026

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More certifications

`databricks-fundamentals.pdf`
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Toolkit

Core	Python, Java, SQL
Machine learning	Scikit-learn, feature engineering, model calibration, NLP
GenAI	RAG, embeddings, hybrid retrieval (BM25 + RRF), cross-encoder re-ranking, LLM evaluation
Data	NumPy, Pandas, Plotly, Matplotlib, Seaborn, Tableau, Excel, Streamlit
Engineering	Git, GitHub Actions, pytest, Docker, FastAPI, Azure, Databricks

DS

Let's build something useful.

I'm looking for a place where I can learn fast and contribute to problems that matter. If that sounds like your team, I'd love to hear from you.

Email me GitHub LinkedIn

Built with care by Diya Shrivastava.
Last updated: November 2026

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Email
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About
Résumé

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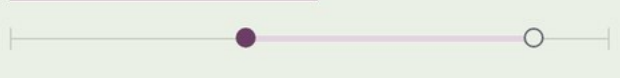

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Work

Projects where I owned the path from raw data to something people could use.

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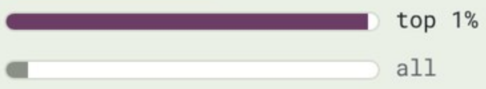

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55% better — than the baseline

Show 3 earlier learning projects

Learning record

Job simulations I completed on Forage to practise real-world data problems.

COMPANY	SIMULATION	DATE	WHAT I DID	
British Airways	Data Science	May 2026	Analysed customer reviews to find what drives sentiment, and modelled the factors behind buying behaviour.	Certificate
BCG	Data Science	Apr 2026	Scoped a customer-churn investigation, tuned a random-forest churn model to 50% recall, and wrote an executive summary.	Certificate
Commonwealth Bank	Intro to Data Science	Apr 2026	Built data pipelines, anonymised personal data for privacy, and designed a database schema.	Certificate
JPMorgan Chase	Quantitative Research	Apr 2026	Estimated customers' probability of default on a loan book, and bucketed FICO scores with dynamic programming.	Certificate
Deloitte	Data Analytics	Apr 2026	Built a Tableau dashboard for a forensic-technology investigation and drew business conclusions from classified data.	Certificate

← All work

Legal & procurement · Sep–Oct 2026

ClauseCheck — Procurement RAG Assistant

An assistant that answers plain-English questions about supplier contracts, backs every answer with the exact clause it came from, checks every figure against that clause — and refuses when the documents don't say.

ROLE	Group project (team of 2). I built the document and retrieval pipeline — sourcing, loading and cleaning, chunking, the evaluation question set, vector and keyword search, fusion and re-ranking — then led integration and debugging, and built versions 2 and 3, including the redesigned interface and figure checking. My teammate built the answer generation, relevance gate, citation validator, API and first frontend.
STACK	Python · ChromaDB · BM25 · Sentence-Transformers · Groq · Gemini · SQLite · FastAPI · Streamlit
CONTEXT	Built during a simulated industry sprint. View code on GitHub →

At a glance

93.5% `recall@5`
Correct clause in the top 5 for 29 of 31 answerable questions

●●●●●●●●●●●●●●●●●●●●●●●●●●●●●○○

9 of 9 `refused`
Every question the contracts can't answer was refused, not guessed

●●●●●●●●●

1 of 31 `wrongly refused`
A 3.2% false-refusal rate on answerable questions

All three come from the 40 tuning questions. The 27 held-out questions haven't been run yet.

Read the limitation ↓

158 `documents`
From 69 suppliers, split into 8,062 clause-level chunks

375 `tests`
Passing in the final version, up from 262 at version 1

1–5 s `per answer`
Typical time to a checked answer

The problem

Procurement teams hold contracts with 100–200 suppliers, but the answer to a simple question — what are our payment terms? — can be spread across a master agreement, a statement of work, later amendments and price schedules. Checking one term by hand takes about ten minutes: find the document, make sure no amendment changed it, then find the exact clause. So people skip it — and invoices get overpaid, early-payment discounts and renewal deadlines get missed, and expired supplier certificates go unnoticed. A generic chatbot doesn't help: in a contract, a confident wrong answer is worse than none.

The data

Mostly real, public documents: 58 commercial contracts from the CUAD dataset (Atticus Project), chosen for procurement relevance; 12 SEC EDGAR amendments paired with their original agreements; and 18 public data-processing and service-level agreements from vendors including Google Cloud, Amazon, Atlassian, Stripe, Twilio and HubSpot. I added 70 synthetic price schedules and compliance certificates, generated with a fixed seed, so every file type (PDF, HTML, Word, text) and clause type was covered.

In total: 158 documents from 69 suppliers. Cleaning stripped headers, footers and redaction marks and standardised dates, so each chunk keeps a stable ID that citations can point to.

Approach

- **Clause-aware chunking.** Contracts are split at clause boundaries, not fixed lengths, so every retrieved piece is a complete clause a citation can point to.
- **Hybrid retrieval.** Meaning-based vector search finds paraphrases; keyword search (BM25) finds exact terms like "Net 45" that embeddings can blur. The two ranked lists are merged with Reciprocal Rank Fusion, then a cross-encoder re-ranks the top results.
- **Local embeddings.** Embeddings run on the machine instead of through an API. With 158 documents this costs nothing and keeps the free-tier model quota for answering questions.
- **Amendments follow their originals.** When an amendment is retrieved, the clause it changes in the original agreement comes with it, and vice versa, up to three links deep.
- **Refuse before guessing.** If the best match scores below a tuned relevance threshold, the system refuses before calling the language model at all — no answer is invented, and no quota is spent.
- **Check every answer.** After an answer is written, code — not the model — confirms that every citation exists in the documents and every number appears in the cited clause. Passing answers get a "Checked" seal; anything else is marked for a person to verify.

Integration and debugging

When the two halves of the system first ran together, 3 of 223 tests failed and generation didn't work at all. I led the integration and kept a debugging log of 37 findings — from installation failures and retired model names to answers refused after an amendment upload — and fixed 24 of them. The rest are recorded as open, partly fixed or by design. The log also records why the relevance threshold was re-tuned three times as real questions exposed where it was too strict.

Evaluation

A labelled set of 67 questions: 30 drawn from real CUAD contracts (renewal, termination, liability, pricing, compliance), 22 exact-value questions (unit prices, payment terms, certificate expiry dates), and 15 questions the documents cannot answer. The set was split into 40 tuning and 27 held-out questions. Measured on the tuning set with real models and caching off: the correct clause was in the top 5 for 29 of 31 answerable questions, all 9 unanswerable questions were refused, and 1 answerable question was wrongly refused.

`fig-checked-answer.png`

Screenshot to add: the app answering "What are the payment terms?" for Orion

A checked answer: the payment period comes from Amendment 1, which replaced the original clause. Every citation was found and every number matched the clause text.

Results

93.5% recall@5, every unanswerable question refused, a 3.2% false-refusal rate, and checked answers in one to five seconds. The final version has a redesigned interface with per-contract conversations, a "Terms found" table showing which clause is in force and since when, and an insights page tracking answer rate and speed against targets.

Limitations

> - **Tuned, not yet held out.** All headline figures come from the 40 tuning questions — the same ones used to set the threshold. The 27 held-out questions haven't been run, so the figures may be optimistic.
> - **Citation accuracy isn't measured.** The checks confirm citations exist and numbers match, but whether the cited clause is the right one hasn't been scored by the evaluation judge yet.
> - **A small evaluation set.** Forty questions is a thin basis for a threshold that decides when to refuse.
> - **Smaller than a real repository.** 158 documents is a simulation; the price schedules and certificates are synthetic, and performance on a large, messy enterprise collection is untested.
> - **Fragile when free models fail.** If both model providers hit rate limits, a request can take over 40 seconds and end in a generic error. There's no retry or circuit breaker yet.
> - **Upload order matters.** An amendment uploaded before its original agreement isn't linked to it.

What I'd do next

1. **Run the held-out set** and re-check the threshold on questions it was never tuned on.
2. **Measure citation accuracy** with the evaluation judge, since it's what users actually trust.
3. **Compare each retriever on its own** — vector only, keyword only and hybrid — to show what each part contributes.
4. **Link amendments automatically** by supplier and date, so upload order stops mattering.
5. **Add retries and clear errors** for when model providers are rate-limited.

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Back to all work

About

I build machine-learning systems that are honest about what they know.

Most of my projects end with a question that isn't about the model: **what will someone actually do with this output?** A dispute-risk score is only useful once it becomes a decision — allow, challenge or review — with a rupee cost attached. A contract assistant is only useful if a procurement team can check every answer against the clause it came from.

That question shapes how I work. I split data the way time actually flows, so a model never learns from the future. I **calibrate** scores so that a 10% risk really means 10%. And I test what a system does when it doesn't know: ClauseCheck **refuses rather than guesses**, and ChargebackLens shows the segments where it **shouldn't be deployed at all**. Some of my most important results are the ones that say no.

I study Computer Science (Software Product Engineering) at Lovely Professional University, where I hold a 9.10 CGPA. I started out building full-stack and mobile apps before moving into machine learning and GenAI. I'm looking for a team that has shipped models into the real world, and where being careful is treated as part of being good.

My path

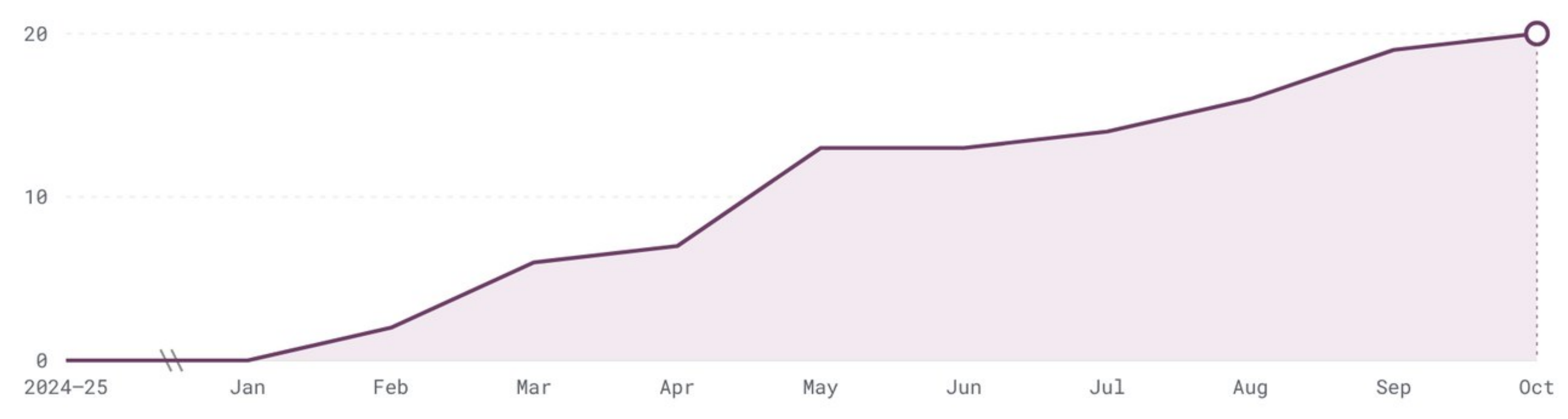
The climb so far, then every step of the trail — newest first.

Elevation gained

Milestones completed since 2024 — projects, certifications, simulations and programs

`20` ▲ `+1 in the last month`

`Now` `1 month` `3 months` `6 months`



Now — ClauseCheck in October — 20 milestones since I began my B.Tech in 2024.

Projects	Certifications	Simulations	Programs
7	7	5	1

`2028`
Expected graduation.

Currently
Building more projects, learning further technologies and looking for an internship.

`Sep–Oct 2026`
Built ClauseCheck as a group project.

`Sep 2026`
Earned Databricks Fundamentals.

`Aug–Sep 2026`
Built ChargebackLens for the Razorpay AI Buildathon, and StreakForge, a fitness retention analytics platform, as a team project.

`Aug 2026`
Earned Microsoft Power Platform PL-900 and PL-200.

`Jul 2026`
Took part in My Equation's 15-day Resident Robotics Program, working with ROS2, Gazebo simulation and Sim2Real methods.

`Jun 2026`
Completed four semesters with a CGPA of 9.10/10.

`Apr–May 2026`
Completed five Forage job simulations (BCG, JPMorgan Chase, Deloitte, Commonwealth Bank, British Airways) and built DeliverIQ.

`Mar–Apr 2026`
Built HealthOps, hospital resource analytics.

`Mar 2026`
Completed Anthropic Academy: AI Fluency and Claude 101.

`Feb–Mar 2026`
Built CraftBridge, a marketplace app for local artisans.

`Feb 2026`
Earned Microsoft Azure AI Fundamentals (AI-900) and GitHub Foundations (GH-900).

`Feb–Mar 2026`
Built ParkEase, a full-stack smart parking platform.

`2024–2026`
Learnt web development and computer-science fundamentals while focusing on my academic courses.

`2024`
Began B.Tech CSE (Software Product Engineering) at Lovely Professional University, batch 2024–2028.

What I care about

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Certifications

CERTIFICATION	ISSUER	DATE	
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Power Platform Functional Consultant Associate (PL-200)	Microsoft	`Aug 2026`	Verify
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Education

B.Tech Computer Science & Engineering (Software Product Engineering)

Lovely Professional University · 2024–2028 · `CGPA 9.10/10`

Relevant courses

Completed with outstanding and excellent grades

CORE COMPUTER SCIENCE

- Data Structures & Algorithms
- Operating Systems
- Database Management Systems
- Discrete Mathematics

PROGRAMMING

- Object-Oriented Programming Lab
- Problem Solving Using Programming

WEB & PRODUCT

- Front-End Web Development
- Back-End Web Development
- Full-Stack Development
- Design for Developers

● STUDYING NOW

- Design & Analysis of Algorithms
- DevOps Fundamentals

Style guide

Light, calm and sincere: pale sage paper, graphite ink, one plum accent, a book serif and lots of space.

Colour

Background	#F5F7F2	Pale sage page
Surface	#FFFFFF	Now card, boxes
Muted band	#EBF0E7	Bands, project cards
Border	#DEE5D8	Hairlines
Accent tint	#F3EAF1	Highlighter, badge
Ink	#23262B	Graphite text
Ink muted	#5D626A	Captions, metadata
Accent	#6C3E66	Plum: links, buttons, metrics
Accent hover	#5A3355	Hover and pressed
Limitation	#8A5A1F	On #FBF1E4 callouts

Type

Display · Literata 600 60 / 40px	**People can rely on**
H2 · Literata 600 32 / 26px	**How I approach a problem.**
Value title · Literata italic 24 / 22px	*Honest numbers.*
Body · Public Sans 400 18 / 17px	A model is only useful when its output is something someone can act on.
Metric · Roboto Mono 600 36 / 32px	**₹11.99L**

Components

See my work

Download résumé

● Open to AI/ML and Data Science internships — available now.

Standalone link (underline on hover)

In a sentence, a link stays underlined.

Tech stack · Python · Scikit-learn · Streamlit

Project card

DeliverIQ

Read case study →

7.83 min

MAE · average prediction error

Limitation callout. Mandatory on every featured case study.

Space

160px between home sections on laptops, 112px on phones.

Width

680px for reading, 1040px for project lists and figures.

Corners

16px on cards and boxes; pill for buttons and the badge.

Motion

Sections fade and rise 8px over 400ms, once. Off under reduced motion.